Ex: 14 Implement packet sniffing using Raw Sockets.

8/10/2025

Aim:

→ To Implement packet sniffing using raw Sockets in Python and capture network packets

Procedure:

1. Open Vs code and create a Python file named Packet_sniffer.py
2. Import the required socket and struct libraries
3. Create and Configure a raw socket to sniff local IP address
4. Run Vs code as Administrator and enter the Program.
5. Generate network traffic using ping 8.8.8.8
6. Capture & display the Source IP, destination IP Protocol and Packet Size.

Result:

The Packet sniffing was successfully using raw Sockets in Python, and network packet Local IP and other details were displayed.

Program:

```
from scapy.all import sniff, IP
def Packet_callback(packet):
    if IP in packet:
        print("Protocol:", packet[IP].proto)
        print("Source IP:", packet[IP].proto)
        print("Destination IP:", packet[IP].dst)
        print("-"*40)
print("Packet Sniffing Started...")
print("Capturing 10 packets...")
sniff(prn=Packet_callback, count=10)
print("Packet Sniffing Completed")
```

Output:

Protocol: 17

Source IP: 172.16.~~19.214~~ ~~254.255.255.250~~

Destination IP: ~~239~~.255.~~255~~.250